MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Guidelines for Inpatient Administration of Nirsevimab to Infants and Children

I. INDICATIONS FOR NIRSEVIMAB INPATIENT ADMINISTRATION - 0 TO 19 MONTHS OF AGE

<table>
<tr><th></th><th>0-7 mos</th><th>8-11 mos</th><th>12-19 mos</th></tr>
<tr><td colspan="4">No maternal RSV vaccine (Abrysvo) administered this pregnancy, RSV vaccination status unknown, or born < 14 days after RSV vaccination</td></tr>
<tr><td>No risk factors listed below for severe RSV disease</td><td>x 1 during 1st RSV season</td><td></td><td></td></tr>
<tr><td>Risk factors for severe RSV disease</td><td colspan="3"></td></tr>
<tr><td>Cystic fibrosis AND ≥ 1 of the following:
▪ Previous hospitalization for pulmonary exacerbation in the first year of life
▪ Abnormalities on chest imaging at clinical baseline
▪ Weight-for-length < 10th percentile</td><td colspan="3" rowspan="4">x 1 during 1st RSV season AND x 1 during 2nd RSV season</td></tr>
<tr><td>American Indian or Alaska Native descent</td></tr>
<tr><td>Chronic lung disease of prematurity requiring ≥ 1 of the following any time in the prior 6 months:
▪ Chronic corticosteroids, diuretics, or supplemental oxygen therapy</td></tr>
<tr><td>Severely immunocompromised (examples):
▪ Bone marrow/solid organ transplant or CAR T-cell therapy during RSV season
▪ Severe combined immunodeficiency (SCID)
▪ Acute myeloid leukemia</td></tr>
<tr><td colspan="4">Received maternal RSV vaccine (Abrysvo) AND born ≥ 14 days after maternal vaccination during current pregnancy</td></tr>
<tr><td>No risk factors listed below for severe RSV disease AND born to immunocompetent parent</td><td></td><td></td><td></td></tr>
<tr><td>No risk factors listed below for severe RSV disease AND born to immunocompromised parent</td><td>See Section VII</td><td></td><td></td></tr>
<tr><td>Risk factors for severe RSV disease</td><td colspan="3"></td></tr>
<tr><td>Cystic fibrosis AND ≥ 1 of the following:
▪ Previous hospitalization for pulmonary exacerbation in the first year of life
▪ Abnormalities on chest imaging at clinical baseline
▪ Weight-for-length < 10th percentile</td><td rowspan="4">See Section VII</td><td colspan="2" rowspan="4">x 1 during 2nd RSV season</td></tr>
<tr><td>American Indian or Alaska Native descent</td></tr>
<tr><td>Chronic lung disease of prematurity requiring ≥ 1 of the following any time in the prior 6 months:
▪ Chronic corticosteroids, diuretics, or supplemental oxygen therapy</td></tr>
<tr><td>Severely immunocompromised (examples):
▪ Bone marrow/solid organ transplant or CAR T-cell therapy during RSV season
▪ Severe combined immunodeficiency (SCID)
▪ Acute myeloid leukemia</td></tr>
</table>

II. NIRSEVIMAB INPATIENT ADMINISTRATION – CARDIAC BYPASS SURGERY

A. Patients who undergo cardiac bypass surgery after an initial dose of nirsevimab should receive a second dose, regardless of previous risk factors, upon hospital discharge if still within the same RSV season.

III. NIRSEVIMAB DOSING

NIRSEVIMAB DOSING	
Infants < 8 months of age	**Infants and children 8 to 23 months of age**
< 5 kg: 50 mg IM x 1 ≥ 5 kg: 100 mg IM x 1	200 mg IM x 1 (administered as two 100 mg injections)
SUPPLEMENTAL DOSE FOLLOWING CARDIAC BYPASS SURGERY	
Patient's first RSV season	**Patient's second RSV season**
< 5 kg: 50 mg IM x 1 ≥ 5 kg AND ≤ 90 days from initial dose: 100 mg IM x 1 > 90 days from initial dose: 50 mg IM x 1	≤ 90 days from initial dose: 200 mg IM x 1 > 90 days from initial dose: 100 mg IM x 1

IV. NIRSEVIMAB TIMING

A. Period of inpatient administration

1. Nirsevimab administration should only occur during or shortly before periods of active RSV circulation.
2. Administration will ***begin October 1st and conclude no later than March 31st*** based on the typical RSV season in Michigan. Administration may end sooner if RSV PCR positivity at Michigan Medicine falls below 3%.
    a. ******Note that for RSV season 2025/2026 nirsevimab administration will conclude April 30, 2026.******
3. Outside of the typical RSV season, nirsevimab administration will begin when RSV PCR positivity at Michigan Medicine is greater than 3% for 2 consecutive weeks and will end when RSV PCR positivity falls below 3%.

B. Timing during admission

1. For healthy newborns, a single dose of nirsevimab should be administered within the first week of life.
2. Patients with extended birth hospitalizations should receive nirsevimab shortly before discharge.
3. Eligible patients who have not previously received nirsevimab and are admitted after birth may receive nirsevimab before discharge.

V. GENERAL NIRSEVIMAB INFORMATION

A. Coadministration with routine childhood vaccines

1. In accordance with the CDC's general best practices for immunizations, simultaneous administration of nirsevimab with age-appropriate vaccines is recommended.
2. In clinical trials, when nirsevimab was administered concomitantly with routine childhood vaccines, the safety and reactogenicity profile of the concomitant regimen was similar to the childhood vaccines administered alone.
3. When concomitantly administered, nirsevimab is not expected to interfere with the immune response to other vaccines.

B. Duration of protection

1. The duration of effect of nirsevimab has been verified in clinical trials at 5 months. Pharmacokinetic data suggests protection against RSV could be as long as one year.

VI. CONSIDERATIONS FOR PATIENTS ALSO ELIGIBLE FOR PALIVIZUMAB

A. Patients previously eligible for palivizumab per 2014 AAP guidelines[4]

1. 2024 AAP Red Book guidelines supersede previous palivizumab guidelines, and some children who were previously eligible for palivizumab are no longer recommended to receive palivizumab or nirsevimab. These include patients 8 to 12 months with significant prematurity,[1] hemodynamically significant heart disease,[2] and certain anatomical abnormalities.[3] (See Footnotes). This also includes patients >19 months with chronic lung disease, cystic fibrosis, or severe immunocompromise.
2. Page Antimicrobial Stewardship (pager: 36149) for discussion of individual cases.

B. Mixed use of palivizumab and nirsevimab in children eligible for nirsevimab

1. Nirsevimab is preferred over palivizumab for protection against RSV infection in nirsevimab-eligible infants. If nirsevimab is not available, palivizumab should be given to infants and children at high risk of severe RSV disease, except if the only risk factor is Native American or Alaska Native heritage.

2. If nirsevimab is administered, palivizumab should not be administered later in the same season.
3. If palivizumab was administered initially for the season and fewer than 5 doses were administered, eligible infants should receive 1 dose of nirsevimab 30 days after the preceding palivizumab dose. No further doses of palivizumab should be administered.
4. If palivizumab was administered in season 1 and the child is eligible for RSV prophylaxis with nirsevimab in season 2, the child should receive nirsevimab in season 2.
5. If nirsevimab is not available, palivizumab should be administered as previously recommended.

VII. CONSIDERATIONS FOR NIRSEVIMAB AND MATERNAL RSV VACCINATION

A. Most infants who are born ≥ 14 days after maternal RSV vaccination in the current pregnancy do not require nirsevimab during their first RSV season. However, nirsevimab can be considered when, per the clinical judgement of the healthcare provider, the potential incremental benefit of administration is warranted, including but not limited to the following circumstances:

1. Infants born to pregnant people who may not mount an adequate immune response to vaccination or have conditions associated with reduced transplacental antibody transfer.
2. Infants who have undergone cardiopulmonary bypass or extracorporeal membrane oxygenation leading to loss of transplacental antibodies.
3. Infants with substantial increased risk for severe RSV disease (e.g., hemodynamically significant congenital heart disease; intensive care admission and requiring oxygen at discharge).

B. Current CDC recommendations make no distinction between those born to pregnant people who had RSV disease during pregnancy and those that did not. These patients were not specifically excluded from trials. Nirsevimab is still indicated in patients born to pregnant people who had RSV infection during pregnancy.

C. If their mother was vaccinated in a prior pregnancy, the infant should receive nirsevimab. CDC continues to evaluate data to determine whether there is sufficient benefit of revaccination in subsequent pregnancies.

D. Data are very limited for these clinical scenarios. Contact Antimicrobial Stewardship with questions (pager # 36149).

VIII. FOOTNOTES

1. Born < 29 weeks gestation
2. Hemodynamically significant congenital heart disease
    a. Pulmonary hypertension requiring ≥ 1 of the following:
        a. Sildenafil, tadalafil, bosentan, ambrisentan, or prostacyclins
    b. Receiving ≥ 1 of the following for pulmonary overcirculation, AV regurgitation, or ventricular dysfunction
        a. Maintenance diuretics, digoxin, or angiotensin converting enzyme inhibitors
3. Anatomical or neuromuscular abnormality impairing ability to clear secretions from upper airway due to insufficient cough

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References

	Citation
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5	Rose EB, Wheatley A, Langley G, Gerber S, Haynes A. Respiratory syncytial virus seasonality—United States, 2014–2-17. *MMWR Morb Mortal Wkly Rep*. 2018;67(2):71-76. DOI: https://doi.org/10.15585/mmwr.mm6702a4
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Revision History

Date	Revision
10/2024	Revised criteria

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source

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